Design and Optimization of High-Power Dual Active Bridge Converter for AI Data Center Applications

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***Abstract*—The rising power demands of modern AI data centers, coupled with the broader push for high power density and efficiency in power-conversion systems, have pushed more research into isolated bidirectional DC-DC converter topologies. This paper presents a detailed design and optimization study of a 25 kW isolated Dual Active Bridge (DAB) DC-DC converter, operating between an 800 V high-voltage primary DC bus and a 48 V low-voltage secondary DC bus which is a voltage architecture that is increasingly being adopted in AI datacenter power delivery networks. The converter is evaluated across a range of switching frequencies (40 to 250 kHz) to assess its performance, focusing on the trade-offs between efficiency and component sizing. Two distinct semiconductor configurations are analyzed: one based on SiC-GaN and the other using SiC-Si. The study includes a comprehensive loss modeling approach. The results reveal that the SiC-GaN configuration provides superior efficiency across the evaluated frequency range. A Pareto analysis is also performed to guide the selection of the best operating point based on efficiency and magnetic component volume. The findings demonstrate the potential of SiC-GaN-based DAB converters for high-performance applications.**

***Index Terms*—Dual Active Bridge, DC-DC converter, SiC, GaN, efficiency, ZVS, power density, semiconductor configuration, loss modeling, energy storage systems, data centers.**

I. Introduction

With AI optimized facilities now routinely demanding tens to hundreds of megawatts of continuous power, meeting this demand efficiently is one of the largest challenges in modern power electronics. Among the available candidates, the dual-active-bridge (DAB) converter has emerged as one of the most promising solutions because of its compact structure, galvanic isolation, and inherent bidirectional power-flow capability. Recent advances in wide-bandgap power devices, particularly SiC and GaN, together with the development of high-performance magnetic materials such as Iron-based nanocrystalline soft magnetic cores have enabled substantial reductions in converter size and weight [1] [2]. As a result, the DAB converter has gained increasing importance [3], [4], [5]. The 800 V-to-48 V conversion architecture studied in this paper aligns closely with emerging datacenter power delivery standards to efficiently distribute power.

The main features that distinguish the DAB converter from other isolated DC–DC converter topologies are: 1) inherent zero-voltage-switching (ZVS) capability, 2) wide voltage-conversion range, 3) natural bidirectional power-flow operation, and 4) high power-transfer capability compared to reduced-switch alternatives [6]. These characteristics make the DAB converter especially suitable for medium- to high-power applications where efficiency, power density, and controllability are critical qualities that are particularly valuable in the context of AI datacenter infrastructure. The conventional DAB topology (Fig. 1) consists of two full-bridge converters linked by a high-frequency transformer and a series commutation inductor (L_k). Power transfer magnitude and direction are regulated by phase-shift modulation between the bridges. ZVS is achieved when the inductive energy in the commutation loop is sufficient to charge and discharge the equivalent output capacitance (C_{oss}) of the commutating devices before the dead-time expires [7]. This is critical for maintaining high efficiency at the elevated switching frequencies.

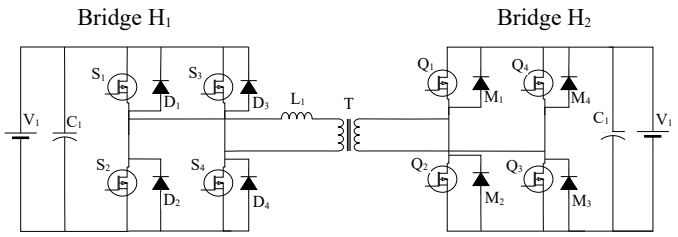


Fig. 1: Schematic of DAB Converter

II. Methodology

A. System Architecture and Device Selection

The proposed system addresses the demands of modern datacenter power distribution by employing an isolated full-bridge Dual Active Bridge (DAB) converter. The architecture consists of a high-voltage primary DC bus (800 V) along with a low-voltage secondary DC bus (48 V). The converter is sized for a nominal continuous power transfer of 25 kW (P_{nom}), incorporating a headroom factor of 1.3 to support transient peak loads [1].

We evaluated two different heterogeneous semiconductor configurations to optimize the trade-off between switching performance and conduction losses:

- **Configuration A (SiC-GaN):** 1200 V SiC primary and 100 V GaN HEMT secondary. Evaluated from 40 kHz to 250 kHz to leverage GaN's zero Q_{rr} and low parasitic capacitance [8]–[10].
- **Configuration B (SiC-Si):** Baseline utilizing a 1200 V SiC primary and 60 V Si secondary, evaluated from 40 kHz to 150 kHz [9], [11].

A device profiling framework integrates static and dynamic parameters such as on-state resistance ($R_{DS(on)}$), parasitic capacitances (C_{iss}, C_{oss}, C_{rss}), gate charge (Q_g), and reverse recovery characteristics (t_{rr}, I_{rm}) directly into the analytical loss model [12]. Galvanic isolation and voltage step-down are achieved via a high-frequency transformer with a turns ratio of $N_p : N_s = 16 : 1$ [13]. Power transfer is governed by Single Phase Shift (SPS) modulation, utilizing the series leakage inductance (L_k) as the primary energy transfer element [12], [14].

B. Modulation Strategy and Component Sizing

Power flow between the primary and secondary bridges is regulated using Single Phase Shift (SPS) modulation [15]. To ensure that the converter can sustain the specified peak transient load ($P_{max} = 32.5$ kW), the equivalent series leakage inductance (L_k) is dynamically sized for maximum power transfer capability at a phase shift ratio of $d = 0.5$ across the frequency sweep:

$$L_k = \frac{V_{in} V_{out}}{8 n f_{sw} P_{max}} \tag{1}$$

where n represents the secondary-to-primary turns ratio (N_s/N_p). Following the inductance size, the DC resistance of the inductor (R_{DC}) is estimated using a geometry constant formulated from typical high-frequency ferrite core (ETD44) parameters and Litz wire characteristics to facilitate accurate passive loss modeling [5].

C. Analytical Loss Modeling

A comprehensive, physics-based analytical loss model evaluates the system across the frequency sweep [16]. The fundamental generalized device-level loss mechanisms comprising switching, conduction, body diode, and gate drive losses are illustrated in Fig. **??** [17]. Note that while Fig. **??** depicts theoretical loss components for a single discrete device, the following formulations aggregate these mechanisms to quantify total dissipation for the high-voltage (HV) and low-voltage (LV) full-bridge architectures, utilizing steady-state currents (I_{rms}, I_{sw}) extracted from the simulation [14], [18].

1) Semiconductor and Passive Component Losses: Conduction losses are evaluated using the device on-state resistance ($R_{DS(on)}$):

$$P_{cond} = 4 \left(I_{rms,h}^2 R_{DS(on),h} + I_{rms,l}^2 R_{DS(on),l} \right) \tag{2}$$

Gate drive losses are calculated as a function of total gate charge (Q_g) and driving voltage ($V_{dr,on}$):

$$P_{gate} = 4 f_{sw} \left(Q_{g,h} V_{dr,on,h} + Q_{g,l} V_{dr,on,l} \right) \tag{3}$$

Passive component losses include the DC copper losses of the commutation inductor ($P_{ind} = I_{rms,in}^2 R_{DC}$) and cabling resistive losses [12].

A Zero Voltage Switching (ZVS) boundary check is performed at each commutation instant. ZVS requires satisfying both an energy constraint (available inductive energy must exceed the equivalent node output capacitance C_{oss} charging energy) and a timing constraint (voltage transition must complete within the allocated dead time t_{dead}) [6]:

$$E_{avail} = \frac{1}{2} L_{comm} I_{sw}^2 > 2 C_{oss} V_{DC}^2 \tag{4}$$

$$t_{trans} = \frac{2 C_{oss} V_{DC}}{I_{sw}} < t_{dead} \tag{5}$$

If both conditions are met, turn-on overlap and diode reverse-recovery losses are eliminated. Otherwise, analytical hard-switching turn-on losses are applied. For Configuration A (SiC-GaN), the LV GaN HEMTs intrinsically lack reverse recovery charge ($Q_{rr} = 0$), significantly reducing these hard-switching penalties [4]. Turn-off transitions are universally treated as hard-switching events, modeled via voltage rise time across the Miller capacitance (C_{rss}) and current fall time governed by C_{iss}.

Dead-time losses occur when bridge current flows through the intrinsic body diode (SiC/Si) or reverse-conduction channel (GaN). The effective conduction interval excludes the switch-node voltage commutation time:

$$t_{dead,eff} = \max \left(t_{dead,actual} - t_{trans}, 0 \right) \tag{6}$$

If ZVS is lost, this interval is assumed to be zero. The equivalent duty ratio for a standard full-bridge is $D_{dead} = 4\, t_{dead,eff}\, f_{sw}$. Total dead-time conduction loss is derived using the effective forward voltage drop (V_f) and instantaneous commutation current (I_{sw}):

$$P_{body} = \left(V_{f,h} \; |I_{sw,h}| \; D_{dead,h} \right) + \left(V_{f,l} \; |I_{sw,l}| \; D_{dead,l} \right) \tag{7}$$

2) Transformer Loss Model: Transformer losses are modeled via a two-term decomposition encompassing magnetic core loss and high-frequency copper loss.

Magnetic core loss density is estimated using an extended Steinmetz empirical relationship [19]:

$$p_{core} = C_{wave}\, k_{fe}\, f_{sw}^{\alpha} \; |B_{pk}|^{\beta} \tag{8}$$

where $C_{wave} = 1.25$ acts as a waveform correction factor for non-sinusoidal square-wave excitation. The Steinmetz coefficient k_{fe} is derived from SIFERRIT N97 material parameters ($K_s = 1.96 \times 10^4$, $\alpha = 1.15$, $\beta = 2.32$) evaluated at a standard reference point [20], [21]. For a full-bridge square-wave excitation of amplitude $\pm V_{in}$, the peak flux density is approximated via Faradays law:

$$B_{pk} = \frac{V_{in}}{4\, N_p\, A_{core}\, f_{sw}} \tag{9}$$

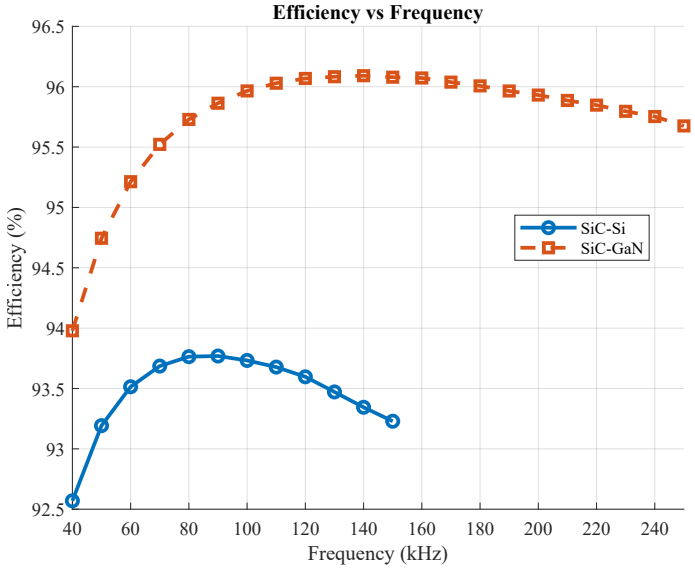


Fig. 2: SiC-GaN vs SiC-Si: Efficiency Variation with Switching Frequency

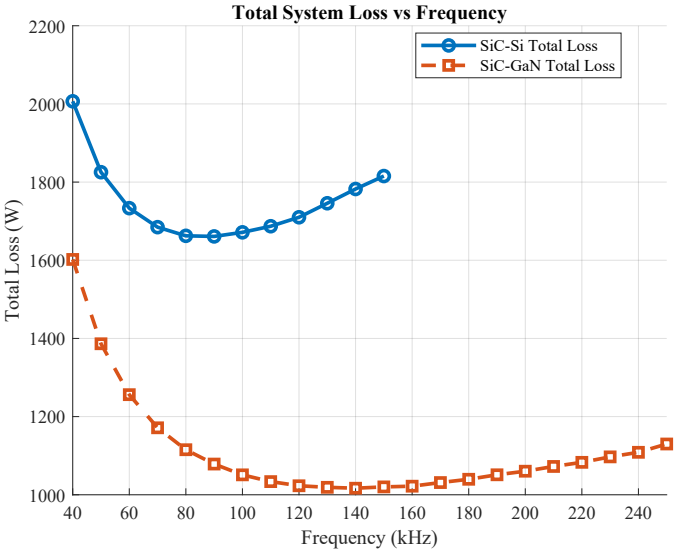


Fig. 3: Comparison of Total System Loss vs. Switching Frequency for SiC-GaN and SiC-Si configurations.

Total core loss is obtained by multiplying the loss density by the physical core volume ($P_{core} = p_{core} V_{core}$).

Copper losses in the primary and secondary windings are computed from the RMS currents and the corresponding AC winding resistances (R_{ac}):

$$P_{cu} = I_{pri,rms}^2 \, R_{ac,pri} + I_{sec,rms}^2 \, R_{ac,sec} \tag{10}$$

The use of AC resistance implicitly accounts for high-frequency skin and proximity effects, with resistance values derived from the Krismer DAB transformer model [4], [12].

3) Inductor and Transformer Sizing: The minimum required volume for the transfer inductor (V_L) is estimated based on the peak magnetic energy storage requirement in the air gap, where $E_L = \frac{1}{2} L_k I_{pk}^2$, yielding $V_L = 2\mu_0 E_L / B_{max,L}^2$. The target peak inductor flux density ($B_{max,L}$) is constrained to 0.30 T.

For the high-frequency transformer, the required effective core cross-sectional area ($A_{e,T}$) is determined by preventing core saturation under square-wave excitation at an allowable flux density ($B_{max,T} = 0.30$ T), expressed as $A_{e,T} = V_{in} / (4 f_{sw} N_p B_{max,T})$. The total transformer volume is subsequently estimated using the empirical scaling law $V_T = k_T A_{e,T}^{1.5}$, where $k_T = 6.0$ serves as a heuristic geometry factor [22].

III. Results and Discussion

A. Efficiency Variation with Switching Frequency

As illustrated in Fig. 2, both configurations exhibit a convex efficiency profile [23]. The SiC-GaN combination achieves a peak efficiency exceeding 96%, whereas the SiC-Si baseline peaks near 93.8% at a slightly lower frequency. Crucially, the SiC-GaN variant maintains superior performance across a substantially wider switching-frequency range.

B. Total Loss and Breakdown Analysis

The variation of total converter loss across the switching-frequency sweep is presented in Fig. 3 [23]. These curves inversely mirror the efficiency trends; total system loss initially decreases to a minimum near the peak efficiency operating point before escalating at higher frequencies [16], [23]. Consistent with previous observations, the SiC-GaN configuration exhibits lower aggregate losses than the SiC-Si baseline across the entire evaluated spectrum [12], [17].

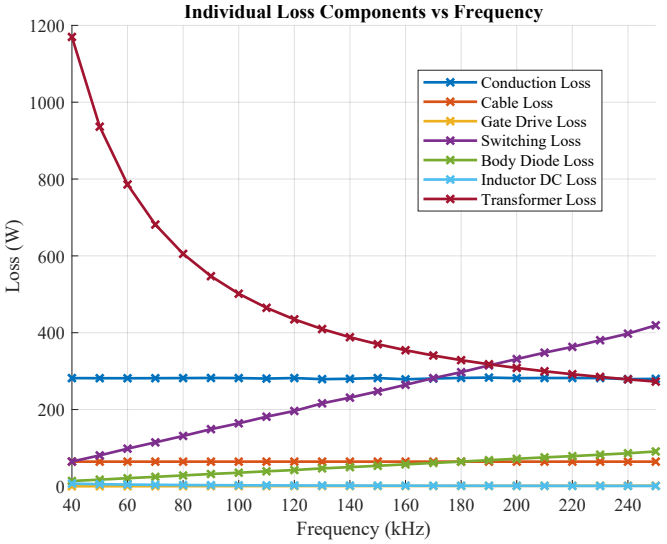
To illustrate the driving factors behind these macro-level trends, Fig. 4 details the individual loss mechanisms. At lower switching frequencies, transformer losses are the dominant source of dissipation for both semiconductor combinations [16]. However, as the operating frequency increases, frequency-dependent mechanisms—specifically switching losses and body diode (or reverse-conduction) losses—scale significantly and eventually dominate the total loss profile [16], [17]. Static dissipation components, such as conduction and inductor DC losses, remain practically invariant across the sweep [12].

C. Magnetic Sizing and System Trade-offs

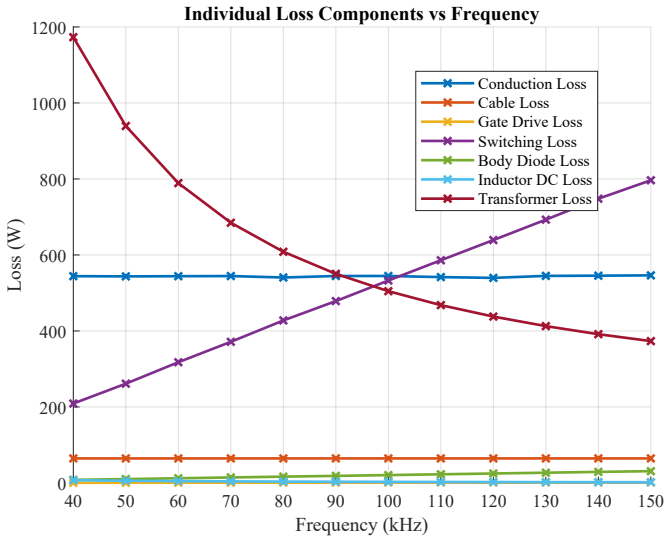
Increasing the switching frequency inherently reduces the required magnetic component volume [2], as depicted in Fig. 5. However, this volumetric benefit must be carefully balanced against the corresponding increase in frequency-dependent losses discussed previously.

To evaluate this system-level trade-off jointly, Pareto fronts mapping total magnetic volume against converter efficiency were constructed at all evaluated operating points (Fig. 6). This representation determines which semiconductor combination offers the most attractive system-level implementation.

The Pareto analysis clearly demonstrates that the SiC-GaN configuration consistently outperforms the SiC-Si baseline,



(a) SiC-GaN Configuration



(b) SiC-Si Configuration

Fig. 4: Loss breakdown analysis highlighting dominant dissipation mechanisms across the switching-frequency sweep.

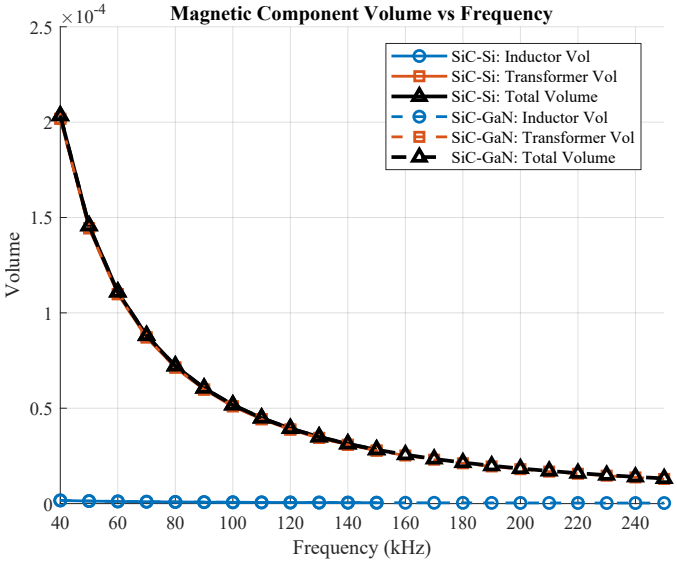


Fig. 5: Comparison of total magnetic component volume vs. switching frequency.

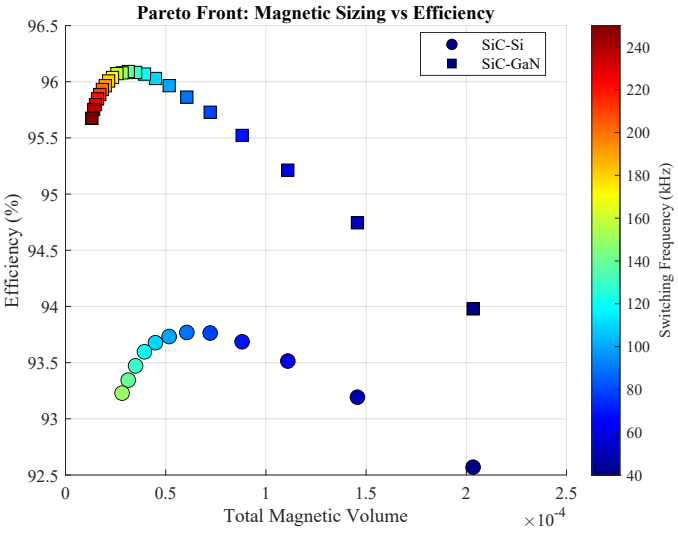


Fig. 6: Pareto trade-off between peak efficiency and total magnetic volume.

offering higher efficiencies for any given magnetic volume constraint. While SiC-Si remains a viable candidate depending on cost or application-specific contexts, SiC-GaN provides a superior wide-frequency operational envelope. Ultimately, the final design choice and operating point selection must be dictated by the specific application's required balance between system efficiency and volumetric footprint.

The relevant waveforms for SiC-GaN at 140 KHz has been shown in Fig. 7

IV. Conclusion

This paper presented a comprehensive system-level evaluation of SiC-GaN and SiC-Si semiconductor configurations in a 25 kW DAB converter. The analytical models and Pareto optimization reveal that the heterogeneous SiC-GaN configuration consistently outperforms the SiC-Si baseline across the 40 kHz to 250 kHz spectrum. Specifically, the GaN-based secondary eliminates reverse-recovery losses and minimizes switching penalties, enabling peak efficiencies exceeding 96% and maintaining superior performance over a wider frequency range. While SiC-Si remains a viable, cost-effective candidate for lower frequency operation, the Pareto analysis demonstrates that SiC-GaN offers a superior trade-off envelope, yielding higher efficiencies for any given magnetic volume constraint.

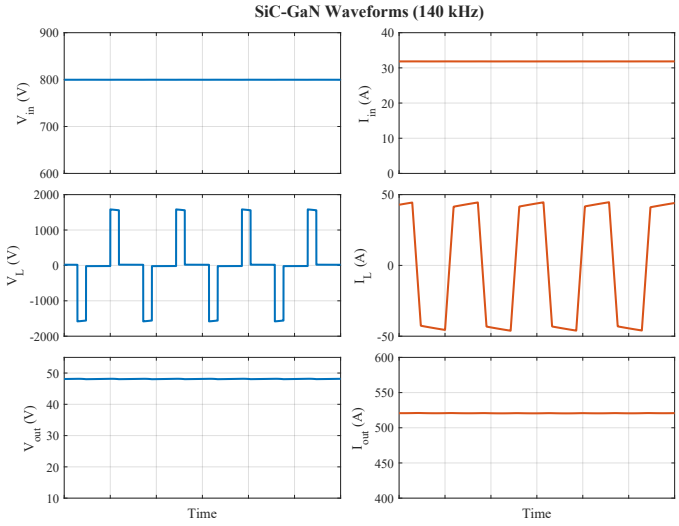


Fig. 7: Converter Steady-State Waveforms for SiC-GaN at 140KHz.

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